



Review

Good practices for estimating area and assessing accuracy of land change



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ABSTRACT

The remote sensing science and application communities have developed increasingly reliable, consistent, and robust approaches for capturing land dynamics to meet a range of information needs. Statistically robust and transparent approaches for assessing accuracy and estimating area of change are critical to ensure the integrity of land change information. We provide practitioners with a set of “good practice” recommendations for designing and implementing an accuracy assessment of a change map and estimating area based on the reference sample data. The good practice recommendations address the three major components: sampling design, response design and analysis. The primary good practice recommendations for assessing accuracy and estimating area are: (i) implement a probability sampling design that is chosen to achieve the priority objectives of accuracy and area estimation while also satisfying practical constraints such as cost and available sources of reference data; (ii) implement a response design protocol that is based on reference data sources that provide sufficient spatial and temporal representation to accurately label each unit in the sample (i.e., the “reference classification” will be considerably more accurate than the map classification being evaluated); (iii) implement an analysis that is consistent with the sampling design and response design protocols; (iv) summarize the accuracy assessment by reporting the estimated error matrix in terms of proportion of area and estimates of overall accuracy, user's accuracy (or commission error), and producer's accuracy (or omission error); (v) estimate area of classes (e.g., types of change such as wetland loss or types of persistence such as stable forest) based on the reference classification of the sample units; (vi) quantify uncertainty by reporting confidence intervals for accuracy and area parameters; (vii) evaluate variability and potential error in the reference classification; and (viii) document deviations from good practice that may substantially affect the results. An example application is provided to illustrate the recommended process.

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1. Introduction

Land change maps quantify a wide range of processes including wildfire (Schroeder, Wulder, Healey, & Moisen, 2011), forest harvest (Olofsson et al., 2011), forest disturbance (Huang et al., 2010), land use pressure (Drummond & Loveland, 2010) and urban expansion (Jeon, Olofsson, & Woodcock, 2013). Map users and producers are acutely interested in communicating and understanding the quality of these maps. Accordingly, guidance on how to assess accuracy of these maps in a consistent and transparent manner is a necessity. The use of remote sensing products depicting change for scientific, management, or policy support activities all require quantitative accuracy statements to buttress the confidence in the information generated and in any subsequent reporting or inferences made. Area estimation, whether of change in land cover/use or of status of land cover/use at a single date, is a natural value-added use of land change maps in many local, national and global land accounting applications. For example, the amount of land area allocated for a specific use is a key country reporting requirement to the United Nations (UN) Food and Agriculture Organization (FAO) statistics and the global forest resources' assessment (FAO, 2010) as well as for countries reporting under the Kyoto protocol and the evolving activities for the UN Collaborative Programme on Reducing Emissions from Deforestation and Forest Degradation – UN-REDD (Grassi, Monni, Federici, Achard, & Mollicone, 2008; UN-REDD, 2008). Estimates of forest extent or deforestation are often derived via remote sensing (cf. Achard et al., 2002; DeFries et al., 2002; Hansen, Stehman, & Potapov, 2010), and area estimation also plays a prominent role in ongoing efforts to establish scientifically valid protocols for forest change monitoring in the context of specific accounting applications to policy approaches for reducing greenhouse gas emissions from forests (DeFries et al., 2007; GOCF-GOLD, 2011).

A key strength of remote sensing is that it enables spatially exhaustive, wall-to-wall coverage of the area of interest. However, as might be expected with any mapping process, the results are rarely perfect. Placing spatially and categorically continuous conditions into discrete classes may result in confusion at the categorical transitions. Error can also result from the change mapping process, the data used, and analyst biases (Foody, 2010). Change detection and mapping approaches using remotely sensed data are increasingly robust, with improvements aimed at the mitigation of these sources of error. However, any map made from remotely sensed data can be assumed to contain some error, with the areas calculated from the map (e.g., pixel counting)

also potentially subject to bias. An accuracy assessment identifies the errors of the classification, and the sample data can be used for estimating both accuracy and area along with the uncertainty of these estimates. While the notion of accuracy assessment is well-established within the remote sensing community (Foody, 2002; Strahler et al., 2006), studies of land change routinely fail to assess the accuracy of the final change maps and few published studies of land change make full use of the information obtained from accuracy assessments (Olofsson, Foody, Stehman, & Woodcock, 2013).

1.1. Good practice recommendations

In this article, we synthesize the current status of key steps and methods that are needed to complete an accuracy assessment of a land change map and to estimate area of land change. This article addresses the fundamental protocols required to produce scientifically rigorous and transparent estimates of accuracy and area. The set of good practice recommendations provides guidelines to assist both scientists and practitioners in the design and implementation of accuracy assessment and area estimation methods applied to land change assessments using remote sensing. The accuracy and area estimation objectives are linked via a map of change. A change map provides a spatially explicit depiction of change and this spatial information can be readily aggregated to calculate the total mapped area or the proportion of mapped area of change for the region of interest (ROI). Accuracy assessment addresses questions related to how well locations of mapped change correspond to actual areas of change. A fundamental premise of the recommended good practices methodology is that the change map will be subject to an accuracy assessment based on a sample of higher quality change information (i.e., the reference classification). The higher quality reference classification is compared to the map classification on a location-specific basis to quantify accuracy of the change map and to estimate area. Although it is possible to estimate area of change without producing a change map (Achard et al., 2002; FAO, 2010; Hansen et al., 2010), we will assume that a map of change exists (although there will not necessarily be a map for each date). The focus for this document is change between two dates.

Before any detailed planning of the response and sampling designs is undertaken, a basic visual assessment should be conducted to identify obvious errors and concerns in the remotely sensed product. This assessment provides an evaluation of the map's suitability for the intended application and should detect if a map is so unsuitable for

use that there is no value in proceeding to a more detailed assessment. The visual assessment should also highlight errors that are easy to remove enabling the map to be refined prior to initiating a detailed assessment or confirm that no obvious concerns exist and the map is ready for further rigorous evaluation.

We separate the accuracy assessment methodology into three major components, the response design, sampling design, and analysis (Stehman & Czaplewski, 1998). The response design encompasses all aspects of the protocol that lead to determining whether the map and reference classifications are in agreement. Because it is often impractical to apply the response design to the entire ROI, a subset of the area is sampled. The sampling design is the protocol for selecting that subset of the ROI. The analysis includes protocols for defining how to quantify accuracy along with the formulas and inference framework for estimating accuracy and area and quantifying uncertainty of these estimates. A separate section of this guidance document is devoted to each of these three major components of accuracy assessment methodology. These sections are followed by an example of the recommended workflow.

1.2. Context of good practice recommendations

The good practice recommendations are intended to represent a synthesis of the current science of accuracy assessment and area estimation. We fully anticipate that improved methods will be developed over time. As the designation of “best practice” implies a singular approach, we prefer the use of “good practice” to indicate that “best” is relative and will vary, with one hard-coded approach not always appropriate. In communicating good practices, desirable features and selection criteria can be followed to ensure that the protocol applied satisfies – as thoroughly as possible – the accuracy and area estimation recommendations. The good practice recommendations do not preclude the existence of other acceptable practices, but instead represent protocols that, if implemented correctly, would ensure scientific credibility of the results. Furthermore, the recommendations presented herein allow flexibility to choose specific details of the different components of the methodology. For example, while the general recommendation for the sampling design is to implement a probability sampling protocol, there are numerous sampling designs that meet this criterion (Stehman, 2009). Similarly, the response design protocol allows flexibility to use a variety of different sources for determining the reference classification and multiple options exist for defining agreement between the map and reference classifications. The good practices recommendations represent an ideal to strive for, but it is likely that most projects will not satisfy every recommendation. Documenting and justifying deviations from good practices are expected features of many accuracy assessment and area estimation studies. For the most part, the good practice recommendations consist of methods for which there is considerable experience of practical use in the remote sensing community.

These good practice recommendations for area estimation and accuracy assessment of land change build on earlier guidelines for single-date land-cover maps described by Strahler et al. (2006). Strahler et al. (2006) presented general guiding principles of good practices with less emphasis on details of methodology. In the intervening years since Strahler et al. (2006), additional theory and practical application related to accuracy assessment and area estimation have been accumulated, and this current document avails upon these developments to delve more deeply into methodological details. We do not attempt to provide an exhaustive description of methods given the range of issues and the highly application-specific nature of the topic. Instead, our purpose is to focus upon the main issues needed to establish a common basis of good practice methodology that will be generally applicable and result in transparent methods and rigorous estimates of accuracy and area. A list of recommendations for all components of the process (sampling design, response design, and analysis) is presented in the Summary section (Section 6).

Estimating area and accuracy of change maps introduces additional methodological challenges that were not within the scope addressed by Strahler et al. (2006). In particular, the area estimation objective was not addressed at all by Strahler et al. (2006). Accuracy assessment of change highlights many unique challenges, including the dynamic nature of the reference data, and aspects of the change features including type, severity, persistence, and area. Another challenge is that change is usually a rare feature over a given landscape. The accuracy of a map and the area estimates derived with its aid are a function of the land-cover mosaic under study, the underlying imagery and the methods applied. Accuracy and area estimates for the same region will, for example, vary if using a per-pixel or object-based classification or if the spatial resolution of the imagery is altered (cf. Baker, Warner, Conley, & McNeil, 2013; Duro, Franklin, & Duba, 2012; Johnson, 2013).

Our recommendations also focus on methods for providing rigorous estimates of land (area) change and its uncertainties. A primary use of such estimates is in analysis and accounting frameworks such as national inventories. In evolving frameworks compensating for successful climate change mitigation actions in the forest sector (such as REDD+, DeFries et al., 2007), the consideration of uncertainties are likely linked with financial incentives and are subject to critical international political negotiations on reporting and verification (Sanz-Sanchez, Herold, & Penman, 2013). Understanding and management of uncertainties in area change is essential, particularly because data and capacity gaps in forest monitoring are large in many developing countries (Romijn, Herold, Kooistra, Murdiyarso, & Verchot, 2012). Accuracy assessments should also focus on identifying and addressing error sources, and prioritize on capacity development needs to provide continuous improvements and reduce uncertainties in the estimates over time. This also includes assessing the value of data streams from evolving monitoring technologies (de Sy et al., 2012; Pratihast, Herold, de Sy, Murdiyarso, & Skutsch, 2013) where the ultimate impact on lower uncertainties need to be proven in operational contexts. Thus, the methods of good practice presented here are generic for providing rigorous estimates, and having agreed-upon tools to do so will provide the saliency and legitimacy for using them in quantifying improvements in monitoring systems, and for dealing with uncertainties in financial compensation schemes (e.g., for climate change mitigation actions).

This article synthesizes key steps and methods needed to complete an accuracy assessment of a change map and to estimate area and accuracy of the map classes. It addresses the protocols required to produce scientifically rigorous and transparent estimates of accuracy and area.

2. Sampling design

The sampling design is the protocol for selecting the subset of spatial units (e.g., pixels or polygons) that will form the basis of the accuracy assessment. Choosing a sampling design requires a consideration of the specific objectives of the accuracy assessment and a prioritized list of desirable design criteria. The most critical recommendation is that the sampling design should be a probability sampling design. An essential element of probability sampling is that randomization is incorporated in the sample selection protocol. Probability sampling is defined in terms of inclusion probabilities, where an inclusion probability relates the likelihood of a given unit being included in the sample (Stehman, 2000). The two conditions defining a probability sample are that the inclusion probability must be known for each unit selected in the sample and the inclusion probability must be greater than zero for all units in the ROI (Stehman, 2001).

A variety of probability sampling designs are applicable to accuracy assessment and area estimation, with the most commonly used designs being simple random, stratified random, and systematic (Stehman, 2009). Non-probability sampling protocols include purposely selecting sample units (e.g., choosing units that are convenient to access), restricting the sample to homogeneous areas, and implementing a complex or ad hoc selection protocol for which it is not possible to derive the

inclusion probabilities. The condition that the inclusion probabilities must be known for the units selected in the sample must be adhered to. These inclusion probabilities are the basis of the estimates of accuracy and area, so if they are not known, the probabilistic basis for design-based inference (see Section 4.2) is forfeited. It is difficult to envision a circumstance in which a deviation from this condition of probability sampling (i.e., known inclusion probabilities) would be acceptable for a scientifically rigorous assessment of accuracy.

In practice, it is not always possible to adhere perfectly to a probability sampling protocol (Stehman, 2001). For example, if the response design specifies field visits to sample locations, it may be too dangerous or too expensive to access some of the sample units. Conversely, persistent cloud coverage or lack of useable imagery for portions of the ROI may prevent obtaining the reference classification for some sample units. The reference data are often derived from another set of imagery and the spatial and temporal coverage of reference data might be different from the coverage of the imagery used to create the map. If the reference classification for a sample unit cannot be obtained, the inclusion probability is zero for that unit. All deviations from the probability sampling protocol should be documented and quantified to the greatest extent possible. For example, the proportion of the selected sample units for which cloud cover prevented assessment of the unit should be reported, or the proportion of area of the ROI for which the reference imagery is not available should be documented. Whereas probability sampling ensures representation of the population via the rigorous probabilistic basis of inference established, when a large proportion of the ROI is not available to be sampled, the question of how well the sample represents the population must be addressed by subjective judgment.

2.1. Choosing the sampling design

The major decisions in choosing a sampling design relate to trade-offs among different designs in terms of advantages to meet specified accuracy objectives and priority desirable design criteria. The objectives commonly specified are to estimate overall accuracy, user's accuracy (or commission error), producer's accuracy (or omission error), and area of each class (e.g., area of each type of land change). Estimates for subregions of the ROI are also often of interest (cf. Scepán, 1999). Desirable sampling design criteria include: probability sampling design, ease and practicality of implementation, cost effectiveness, representative spatial distribution across the ROI, small standard errors in the accuracy and area estimates, ease of accommodating a change in any step in the implementation of the design, and availability of an approximately unbiased estimator of variance. Determining whether any or all of these desirable design criteria have been satisfied by the chosen sampling design may be subjective. For example, determining what constitutes a small standard error will depend on the application and may vary for different estimates within the same project. There are also precedents for defining an accuracy target and desired error bounds as a means for determination of sample size using standard statistical theory (Wulder, Franklin, White, Linke, & Magnussen, 2006) (see also Section 5.1.1).

Stehman and Foody (2009) provide an overview and comparison of the basic sampling designs typically applied to accuracy assessment. Stehman (2009) provides a more expansive review of sampling design options and discusses how these designs fulfill different objectives and desirable design criteria. A variety of sampling designs will satisfy good practice guidelines so the key is to choose a design well suited for a given application. Three key decisions that strongly influence the choice of sampling design are whether to use strata, whether to use clusters, and whether to implement a systematic or simple random selection protocol (Stehman, 2009). Each of these decisions will be discussed in the following subsections.

2.1.1. Strata

There is often a desire to partition the ROI into discrete, mutually exclusive subsets or strata (e.g., a global map could be stratified

geographically by continents). Stratification is a partitioning of the ROI in which each assessment unit is assigned to a single stratum. The two most common attributes used to construct strata are the classes determined from the map and geographic subregions within the ROI. Stratification is implemented for two primary purposes. The first purpose is when the strata are of interest for reporting results (e.g., accuracy and area are reported by land-cover class or by geographic subregion). The second use of stratification is to improve the precision of the accuracy and area estimates. For example, when strata are created for the objective of reporting accuracy by strata, the stratified design allows specifying a sample size for each stratum to ensure that a precise estimate is obtained for each stratum. Land change often occupies a small proportion of the landscape, so a change stratum can be identified and the sample size allocated to this stratum can be large enough to produce a small standard error for the change user's accuracy estimate.

The practical reality is that limited resources will likely be available for the reference sample and this constraint will strongly impact sample allocation decisions because different allocations favor different estimation objectives. For example, allocating equal sample sizes to all strata favors estimation of user's accuracy over estimation of overall and producer's accuracies (Stehman, 2012). Conversely, the standard errors for estimating producer's and overall accuracies are typically smaller for proportional allocation (i.e., the sample size allocated to each stratum is proportional to the area of the stratum) relative to equal allocation. As a compromise between favoring user's versus producer's and overall accuracies, the allocation recommended is to shift the allocation slightly away from proportional allocation by increasing the sample size in the rarer classes, but the sample size for the rare classes should not be increased to the point where the final allocation is equal to allocation (see Section 5 for an example). The sample size allocation decision can be informed by calculating the anticipated standard errors (see Sections 4.3 and 4.4) for different sample sizes and different allocations. An ineffective allocation of sample size to strata will not result in biased estimators of accuracy or area, but it may result in larger standard errors (see Section 5 for an example).

When stratified sampling is applied to a single date land-cover map, it is usually feasible to define a stratum for each land-cover class (Wulder, White, Magnussen, & McDonald, 2007). Identifying an effective stratification for change can be more challenging. A common approach is to use a map of change to identify the strata, and such strata are effective for estimating user's accuracy of change precisely. However, the number of different types of change may be so large that defining every change type as a stratum is not advisable. For example, in a post-classification comparison of two land-cover maps that each include 8 land-cover classes, there are 56 possible types of change in the final change map. If each stratum must receive a relatively large sample to achieve a precise user's accuracy estimate, the overall sample size may be unaffordable.

The trade-offs between precision of user's accuracy, producer's accuracy, and area estimates from different sample size allocations become exacerbated as the number of strata increases. Some types of change may be very unlikely to occur and consequently could be eliminated as strata. To further reduce the number of strata, strata could be defined on the basis of generalized change categories (Wickham et al., 2013). For example, a stratum could be change from any class to urban (i.e., urban gain), and another stratum could be change to any class from forest (i.e., forest loss). These generalized or aggregated change strata are obviously less focused on all possible individual change types. For example, the forest loss stratum could include forest to developed, forest to water, or forest to cropland. These generalized change strata would allow for specifying the sample size allocated to different general change types, but within one of the generalized strata, the sample size allocated to the individual change types would be proportional to the area of that change type. For example, if the most common type of forest loss is to cropland and the least common change is forest loss to water, many more of the sample units within the forest loss stratum will be forest-to-cropland-conversion. Strahler et al. (2006, Fig. 5.2, p. 32) provides

additional examples of aggregated change classes that could be used as strata.

The desire to limit the number of strata motivates discussion of subpopulation estimation as it relates to sampling design. A subpopulation is any subset of the ROI, for example a particular type of change or a particular subregion. Subpopulations can be defined as strata, but it is not necessary for a subpopulation to be defined as a stratum to produce an estimate for that subpopulation. For example, when aggregating multiple types of change into a generalized change stratum, it would still be possible to estimate accuracy of each of the subpopulations representing the individual types of change making up the aggregated change stratum. However, if these subpopulations are not defined as strata, the sample size representing the subpopulation may not be large enough to obtain a precise estimate. Resources available for accuracy assessment may require limiting the number of strata used in the design, so prioritizing subpopulations may be necessary to establish which subpopulations are defined as strata.

It is sometimes the case that several maps will be assessed based on a common accuracy assessment sample. This forces a decision on whether the strata should be based on a single map (and if so, which map) or if the strata should be defined by a combination of the multiple maps. Once strata are defined and the sample is selected using these strata, the strata become a fixed feature of the design because the analysis is dependent on the estimation weight associated with each sample unit and this weight is determined by the sampling design. Fortunately, whatever the decision is to define strata when multiple maps are to be assessed, the sample reference data are still valid to assess any of the maps, even if the strata are defined on the basis of a single map. The principles of estimation outlined in the Analysis section (Section 4) must be adhered to, and this simply requires using the estimation weights for the sample units determined by the original stratified selection protocol. The impact of the choice of strata will be reflected in the standard errors of the estimates. Olofsson et al. (2012) and Stehman, Olofsson, Woodcock, Herold, and Friedl (2012) discuss sampling design issues associated with constructing a reference validation database that would allow assessment of multiple maps.

To summarize the recommendations related to the important question of whether to incorporate stratification in the sampling design, stratifying by mapped change and by subregions is justified to achieve the objective of precise class-specific accuracy and to report accuracy by subregion. If the overall sample size is not adequate to support both class-specific and subregion accuracy estimates, the subregional stratification may be omitted and accuracy by subregion relegated to the status of subpopulation estimation. The recommended allocation of sample size to the strata defined by the map classes is to increase the sample size for the rarer classes making the sample size per stratum more equitable than what would result from proportional allocation, but not pushing to the point of equal allocation. The rationale for this recommendation is that user's accuracy is often a priority objective and we can control the precision of the user's accuracy estimates by the choice of sample allocation. However, the trade-off is that a design allocation chosen solely for the objective of user's accuracy precision (i.e., equal allocation) may be detrimental to precision of estimates of overall accuracy, producer's accuracy, and area, so a compromise allocation is in order. Lastly, defining aggregations of change types as strata may be necessary if the number of strata needs to be limited, and accuracy and area estimates for the individual change types would be obtained as subpopulation estimates.

2.1.2. Cluster sampling

A cluster is a sampling unit that consists of one or more of the basic assessment units specified by the response design. For example, a cluster could be a 3×3 block of 9 pixels or a $1 \text{ km} \times 1 \text{ km}$ cluster containing 100 1 ha assessment units. In cluster sampling, a sample of clusters is selected and the spatial units within each cluster are therefore selected as a group rather than selected as individual entities. Each of the spatial

units within a cluster is still interpreted as a separate unit even though it is selected into the sample as part of a cluster. For example, a 3×3 pixel cluster would require obtaining the reference classification for individual pixels within the cluster.

The primary motivation for cluster sampling is to reduce the cost of data collection. For example, if field visits are required to obtain the reference classification, transit time and costs may be reduced if the sample units are grouped spatially into clusters. Zimmerman et al. (2013) used cluster sampling to reduce the number of raster images (i.e., clusters) required because the primary cost of the sampling protocol was associated with processing the very high resolution images used for reference data. As another example, Stehman and Selkowitz (2010) used a $27 \text{ km} \times 27 \text{ km}$ cluster sampling unit to constrain sample locations to a single day of flight time per cluster when the reference data were collected by aircraft. Cluster sampling may also be motivated by the objectives of an accuracy assessment. For example, a cluster sampling unit becomes necessary to assess accuracy at multiple spatial supports (e.g., single pixel, 1 ha unit, and 1 km^2 unit).

The cost savings gained by cluster sampling should be substantial before choosing this design because the correlation among units within a cluster (i.e., intraclass correlation) often reduces precision relative to a simple random sample of equal size. Focusing on the specific example of estimating land-cover area in Europe, Gallego (2012) showed that a $10 \text{ km} \times 10 \text{ km}$ sampling unit produced equivalent information to that of a simple random sample of only 25 points or fewer. The low yield of information per cluster diminishes the cost advantage of cluster sampling if the intraclass correlation is high. Another potential disadvantage of cluster sampling is that it complicates stratification when the strata are the map classes and the assessment unit is a pixel. In the simplest setting, each cluster would be assigned to a stratum, but rules have to be established for assigning a cluster to a stratum when the cluster includes area of several different classes. Cluster sampling can be combined with stratification of pixels by the map class of each pixel in a two-stage stratified cluster sampling approach (Stehman, Sohl & Loveland, 2003; Stehman, Wickham, Wade & Smith, 2008), but such designs require more complex analysis and implementation protocols than what are required of a stratified design without clusters. Because of the added complexity cluster sampling introduces for sampling design (e.g., accommodating stratification within a cluster sampling design) and estimation (e.g., estimating standard errors), we recommend this design only in cases for which the objectives require a cluster sampling unit or in which the cost savings or practical advantages of cluster sampling are substantial.

2.1.3. Systematic vs. random selection

The two most common selection protocols implemented in accuracy assessment are simple random and systematic sampling (we define "systematic" as selecting a starting point at random with equal probability and then sampling with a fixed distance between sample locations). Both protocols can be implemented to select units from within strata or to select clusters, and both can be applied to a ROI that is not partitioned into strata or clusters. Unbiased estimators of the various accuracy parameters are available from either systematic or simple random selection, so the bias criterion is not a basis for choosing between these options. Instead, the choice of simple random versus systematic depends on how each selection protocol satisfies the priority desirable design criteria (Stehman, 2009). For example, systematic sampling is often simpler to implement when the response design is based on field visits, but the greater convenience of systematic versus simple random is diminished when working with imagery or aerial photographs as a source of the reference data. Typically, systematic selection will yield more precise estimates than simple random selection, but systematic sampling requires use of a variance approximation so if unbiased variance estimation is a priority criterion, simple random is preferred. Simple random selection also is advantageous if it is likely that the sample size will need to be modified during the course of the accuracy

assessment (Stehman et al., 2012). A scenario in which systematic selection opportunistically arises is when accuracy assessment reference data can be simultaneously obtained in conjunction with another field sampling activity. For example, many national forest inventories employ a systematic sample of field plots (Tomppo, Gschwantner, Lawrence, & McRoberts, 2010) and these field plot data may be an inexpensive, high quality source of reference data. In general, the simple random selection protocol will better satisfy the desirable design criteria and is the recommended option. However, systematic selection is also nearly always acceptable.

2.2. A recommended good practice sampling design

Stratified random sampling is a practical design that satisfies the basic accuracy assessment objectives and most of the desirable design criteria. Stratified random sampling affords the option to increase the sample size in classes that occupy a small proportion of area to reduce the standard errors of the class-specific accuracy estimates for these rare classes. Thus this design addresses the key objective of estimating class-specific accuracy. In regard to the desirable design criteria, stratified random sampling is a probability sampling design and it is one of the easier designs to implement. Stratified sampling is commonly used in accuracy assessment so it has an advantage of being familiar to the remote sensing community (cf. Cakir, Khorram, & Nelson, 2006; Huang et al., 2010; Mayaux et al., 2006; Olofsson et al., 2011). Increasing or decreasing the sample size after the data collection has begun is readily accommodated by stratified random sampling, and unbiased variance estimators are available thus avoiding the need to use variance approximations. An assumption implicit in this recommendation is that change between two dates is of interest. Little work has been done to investigate the effective use of strata for multiple change periods. In the case of stratification based on a change map, it is assumed that reference data for the sampled locations exists for the initial date of the change period (e.g., archived imagery or aerial photography is available). If the reference data must be obtained in real time (e.g., via ground visit), it would not be possible to stratify by a change map that does not yet exist at the initial date. An alternative would be to stratify by anticipated change or predicted change, with the effectiveness of such strata dependent on how well the predicted change matched with the ensuing reality of change.

3. Response design

For the accuracy assessment objective, the response design encompasses all steps of the protocol that lead to a decision regarding agreement of the reference and map classifications. For area estimation, the response design provides the best available classification of change for each spatial unit sampled. The four major features of the response design are the spatial unit, the source or sources of information used to determine the reference classification, the labeling protocol for the reference classification, and a definition of agreement. Each of these major features is discussed in the following subsections.

3.1. Spatial assessment unit

The spatial unit that serves as the basis for the location-specific comparison of the reference classification and map classification can be a pixel, polygon (or segment), or block (Stehman & Wickham, 2011). The ROI is partitioned based on the chosen spatial unit (i.e., the region is completely tiled by these non-overlapping spatial units). Commonly, the pixel is selected as the spatial unit. The pixel is an arbitrary unit defined mainly by the properties of the sensing system used to acquire the remotely sensed data or a function of the grid used to sub-divide space in a raster based data set. A polygon is defined as a unit of area, perhaps irregular in shape, representing a meaningful feature of land cover. For example, a polygon may be delineated from a map such that

the area within the polygon has the same map classification (e.g., the entire polygon is stable forest or the entire polygon represents an area of change from forest to urban). Polygons defined on the basis of a map will be called “map polygons.” Alternatively, a polygon could be delineated on the basis of the reference classification as an area within which the reference class is the same. A polygon delineated on the basis of the reference classification will be called a “reference polygon”. A “block” spatial assessment unit is defined as a rectangular array of pixels (e.g., a 3×3 block of pixels). Irrespective of the spatial unit selected, it is important to note that some spatial units may be impure, i.e., they represent an area of more than one class. Mixed pixels are common, especially in coarse spatial resolution data. Similarly, it is possible that a map polygon is not internally homogeneous in terms of the reference classification, and a reference polygon may not be internally homogeneous in terms of the map classification. A polygon defined by a segmentation algorithm would not necessarily be homogeneous in terms of either the map or the reference classifications.

Pixels, polygons, or blocks can be used as the spatial unit in accuracy assessment. Regardless of the unit chosen, a critical feature of the response design protocol is that the spatially explicit character of the accuracy assessment must be retained. Practitioners should aim to have reference data with an equal or finer level of detail than the data used to create the map, but we make no recommendation regarding the choice of spatial assessment unit. However, once the spatial assessment unit has been chosen, there will be good practice recommendations associated with that specific unit and the choice of spatial unit also has implications on the sampling design (Stehman & Wickham, 2011) and analysis. Estimates of accuracy and area derived from the same map but through the use of different spatial units may be unequal.

3.2. Sources of reference data

The reference classification can be determined from a variety of sources ranging from actual ground visits to the sample locations or the use of aerial photography or satellite imagery. There are two ways to ensure that the reference classification is of higher quality than the map classification: 1) the reference source has to be of higher quality than what was used to create the map classification, and 2) if using the same source material for both the map and reference classifications, the process to create the reference classification has to be more accurate than the process used to create the classification being evaluated. For example, if Landsat imagery is used to create the map and Landsat is the only available imagery for the accuracy assessment, then the process for obtaining the reference classification has to be more accurate than the process for obtaining the map classification. Additionally, other spatial data may be used to improve the quality of the reference classification, such as forest inventory data or some form of vector data (e.g., roads, pipelines, or crop records). In this subsection, different potential sources of reference data for assessing accuracy of change are identified and strengths and weaknesses of these sources are described.

Possible reference data sources include field plots, aerial photography, forest inventory data, airborne video, lidar, and satellite imagery (Table 1). Additional sources of freely accessible reference data may also be opportunistically available from data mining and crowdsourcing (Foody & Boyd, 2013; Iwao, Nishida, Kinoshita, & Yamagata, 2006).

Table 1
Possible reference data sources.

Reference data source	Exemplar citation
Field plots	Hyypä et al. (2000)
Air photography	Skirvin et al. (2004)
Forest inventory data	McRoberts (2011); Wulder, White, et al. (2006)
Airborne video	Wulder et al. (2007)
Lidar	Lindberg, Olofsson, Holmgren, and Olsson (2012)
Satellite imagery	Scepan (1999); Cohen et al. (2010)
Crowdsourcing	Iwao et al. (2006); Foody and Boyd (2013)

Practical considerations regarding costs often influence the selection of reference data, or the use of existing data. While existing or lower cost data may be desirable from a purchase perspective, the use of disparate data sources will result in additional effort by project analysts to deal with exceptions and inconsistencies. A key to using disparate data sources is to have the reference data that are actually used in the accuracy assessment be, as much as possible, invariant to source. For example, the creation of attributed change polygons makes the polygon the common denominator, rather than the source data. Creating polygonal change units in a portable format and populating a minimum set of fields to support a consistent labeling protocol is desirable. The information to be recorded for each change unit is itemized in Table 2.

Ideally a data source is available for the entire ROI, representing the change types and dates of interest, at a low cost. The realities versus the ideal result in a series of considerations that are detailed in Table 3. For instance, if the ROI is small, cost may be less of an issue and access may not be relevant. For large area projects over poorly monitored areas, existing data sources are not often available so data purchase and interpretation costs become the dominant criteria. The ease of interpretation and consistency of source reference data permits economies in the project flow for the analysts and also promotes automation of repeated activities. Further, the development of a well-documented and consistent change validation data set will have utility for multiple projects and purposes.

Both high- and very high spatial resolution satellite data are viable candidates for reference data. Imagery is typically considered as very high spatial resolution (VHSR) with a spatial resolution of <1 m and high spatial resolution (HSR) with a spatial resolution of <10 m. Both data sources provide information that is finer than the data used in most large area monitoring projects, which would typically have a spatial resolution of greater than 10 m. At the fine spatial resolution of satellite-borne VHSR imagery, panchromatic is often the only spectral information collected. The typical 400 to 900 nm panchromatic data with small pixels (0.50 m in the case of WorldView-1) closely resemble large scale aerial photography and can be interpreted using established aerial photograph interpretation techniques (Wulder, White, Hay, & Castilla, 2008) or subject to digital analyses (cf. Falkowski, Wulder, White, & Gillis, 2009). Both the SPOT Image® and DigitalGlobe® archives can be accessed through Google Earth™, with the image extents by year portrayed. The presence of freely accessible high spatial resolution imagery online through Google Earth™ also presents low cost interpretation options. Limitations of this approach include a lack of data prior to the initiation of the high spatial resolution satellite commercial era (circa 2000), spatial distribution of available imagery, and the actual temporal revisit of the images available. The reported temporal revisit can be on the order of days based upon an ability to point the sensor head. For instance, IKONOS has off-nadir revisit of 3 to 5 days, with 144 days

required for nadir revisit (Wulder, White, Coops, & Butson, 2008). The implication is that when the sun-surface-sensor viewing geometry changes the structure captured changes, such that trees evident on one image may be occluded in another. For a given on-line accessible source of satellite imagery, it should not be expected that historical, archival, global coverage from launch to present exist. Regardless, the ability to view images from multiple years can help determine that date when a change (e.g., a disturbance) occurred. The additional context provided around particular change events aids with interpretation of change type (e.g., determination of harvesting versus forest removal in support of agricultural expansion).

There are few, if any, reference data sources that are available with a uniform likelihood globally. There are some archival datasets with wide global coverage (e.g., Komsat); although, the utility of these data sets may be limited. The utility of any given reference data source when used to capture and relate change is the date or represented by the data. While less of an issue with satellite data, air photos and maps may not be of a known vintage. Acquisition dates of historic photos are often lost, plus maps are often representative of a period, not a singular date. Knowing the conditions that previously existed may not be helpful if the date of change occurrence is not known.

Over some regions, land use change and silvicultural records may also be available to inform on the land-cover change. Note that forest harvesting is a land-cover change relating a successional stage, rather than a land use change (which implies a permanent change in how a particular parcel of land is used — e.g., forestry to agriculture). This distinction is important for both monitoring and reporting purposes as the permanent removal of forests has differing carbon consequences than forest harvesting (Kurz, 2010).

While the good practice guidelines advocate for use of reference data of finer spatial resolution than the map product, this is especially so for single date interpretations of the reference data. Following the opening of the Landsat archive by the USGS (Woodcock et al., 2008), time series of imagery created new opportunities for using imagery of the same spatial resolution (e.g., Landsat) when archival data are available. Simple visual approaches may be applied, such as in Fig. 1, where a change event (fire) that is evident in 2010 can be timed quite precisely by the evidence captured (smoke plume) showing when the fire occurred. This type of change dating is rather opportunistic and not to be commonly expected.

A more reliable means for determining the timing of change events can be from developing and interrogating time series of images (Kennedy, Yang, & Cohen, 2010). To ensure the quality of time series transitions developed, Cohen, Yang, and Kennedy (2010) created a logic and tool for determining the timing and nature of changes captured (TimeSync, <http://timesync.forestry.oregonstate.edu/>). Based upon the image collection and archiving protocols present through the history of Landsat, the spatial and temporal coverage of imagery is not uniform. The temporal precision possible for dating changes based upon time series analysis is likely weaker for locations that already have a paucity of data. This situation is due to the historic practices followed at given Landsat receiving stations through to the commercial era (during the 1980s) when fewer images were collected and archived (Wulder, Masek, Cohen, Loveland, & Woodcock, 2012). It should not be assumed that the temporal density possible for the conterminous United States is possible for all other regions (Schroeder et al., 2011).

Another critical aspect of the response design is that the change period represented by the reference classification must be synchronous with the change period of the classification. Consider a map representing change between 2000 and 2010. To capture the northern hemisphere peak photosynthetic period, the imagery used for this hypothetical project was collected July 15, 2000, and 10 years later, July 15, 2010. The reference data should be collected in 2010, but ideally not after July 15 (assuming similar satellite overpass times) to avoid confusion. Data collected after July 15, 2010 will have to be vetted to ensure that the change present in the reference data did not occur

Table 2

Example characteristics to record for each change polygon. Some attributes can be generated in the GIS; others will need to be entered by the analyst. Notion is that information is captured and carried to provide insights and a record regarding the changes captured. The aim is that the change polygons can be used in a manner that is invariant to source, but that metadata is captured to explain or better understand any data related anomalies that may emerge.

Attribute	Definition/comments
Change area	Area changed, e.g., polygon size in hectares
Change perimeter	Perimeter of polygon, in meters
Change type	Notation of change type, harvest, fire, insect, urban expansion, agricultural development
Change date	As possible, note the change date. May be available from other records, e.g., when a fire occurred, or the acquisition date of the image or photography used.
Data source	Note the data source from which the change polygon is made
Analyst	Name or code to denote the interpreter
Date interpreted	Note the date when the interpretation occurred

Table 3
Elements for consideration when selecting reference data.

Element	Considerations
Cost	What is the budget? What amount per unit of reference data can be purchased? Is the interpretation/labeling protocol efficient?
Ease of access	Varies by data type. Can field visits be made? Is archival image data available?
Ease of use	Is the data produced in a consistent fashion? Is it in formats that are commonly used?
Opportunity for consistency	Can protocols be developed and applied in a systematic and repetitive fashion? Can some tasks be automated?
Vintage — temporal representation	Is the data representative of a time or time period that is relevant to the change product under consideration?
Spatial coverage	Are there opportunities for multiple reference sites from a given reference data source?
Interpretability of change types	Does the data source capture and portray the change types of interest? E.g., is the spatial resolution sufficiently fine to enable interpretation?
Geolocation	Can the candidate reference data source be assumed to be accurately positioned? Will additional geolocation activities be required?

after the product date of the change map. Imagery from the same year is desired but may not always be possible. As such, it is required that the change reference data approximates the date the change occurred as precisely as available. Multiple images help refine the timing of the change event. Mismatched change periods between the map and reference classifications would be a major source of reference data error.

3.3. Reference labeling protocol

The labeling protocol refers to the steps in the response design that take the information provided by the reference data and convert that information to the label or labels constituting the reference classification. Labeling is far from trivial with numerous definitions for land-cover classes in use (cf. [Comber, Wadsworth, & Fisher, 2008](#)) although recent developments such as the FAO's Land Cover Classification system (LCCS) may act to enhance interoperability ([Ahlqvist, 2008](#)). The labeling protocol should also include specification of a minimum mapping unit (MMU) for the reference classification. The MMU can have important implications for accuracy assessment and area estimation. For example, increasing the size of the MMU will lead to a reduction in the representation of classes that occupy small, often fragmented, patches ([Saura, 2002](#)). Changing the MMU can also impact accuracy estimates, although the effect is most apparent when a large change is made ([Knight & Lunetta, 2003](#)). Small patches present a challenge to mapping (cf. [He, Franklin, Guo, & Stenhouse, 2011](#)) and the accuracy of their mapping will degrade as the MMU is increased. However, it is possible that overall map accuracy may increase with a larger MMU, making it important to ensure that attention is focused on an appropriate measure of accuracy for the application in-hand. The precise effects of the MMU will vary as a function of the land-cover mosaic under study and the imagery used. The MMU specified for the response design does not necessarily have to match the MMU specified for the map. In fact, if the reference classification is intended to apply to a variety of maps, it would be likely that the MMU of the reference classification does not

match the map classification for all maps that might be assessed. Often the reference imagery or information will permit distinguishing smaller patches or features that can be distinguished from the map so a smaller MMU will be possible for the reference classification.

The easiest case for the labeling protocol occurs when the assessment unit is homogeneous and a single reference class label can be assigned (the reference class could be a type of change). Often, however, the situation will be more complex making class labeling less certain. For example, the assessment unit may contain a mixture of classes, and even if the unit is homogeneous, it may be difficult to assign a single label (e.g., change type) because the unit is not unambiguously one of the classes in the legend but instead falls between two of the discrete class options in the legend (i.e., land-cover classes are a continuum represented on a discrete scale). A variety of options exist for labeling a unit when a single reference label does not adequately represent the uncertainty of a unit. One or more alternate reference class labels can be assigned to account for ambiguity in the reference classification. Another option when defining agreement is to construct a weighted agreement based on how closely the different classes are related. For example, in the GlobCover assessment, a “matrix” of class relationships was established ([Mayaux et al., 2006](#), GLC2000). A fuzzy reference labeling protocol may also be employed, such as the linguistic scale devised by [Gopal and Woodcock \(1994\)](#) or a fuzzy membership vector in which the reference label for a unit specifies a membership value for each class ([Binaghi, Brivio, Ghezzi, & Rampini, 1999](#); [Foody, 1996](#)). Another option for mixed units is to specify the proportion of area of each class present in the unit ([Foody, Campbell, Trodd, & Wood, 1992](#); [Lewis & Brown, 2001](#)). A different characterization of uncertainty in the reference classification is obtained by assigning a confidence rating that represents the interpreter's perception of uncertainty in the reference classification for that unit. For example, low, moderate and high confidence ratings would indicate increasing confidence on the part of the interpreter that the reference classification is correct. Typically this information can then be used in the analysis to subset results by



Fig. 1. Landsat data can be used for the visual dating of change, with the fire event in progress in inset A, August 3, 2010, with the burned forest outcome evident in inset B, September 20, 2010, Yukon, Canada (Landsat Path 55, Row 18).

photography can illuminate subtle changes in forest conditions such as decline due to insects or water stress and converse recovery of forests following disturbance. The response design protocols presented also do not address the situation in which the map provides information as a continuous variable. Although many of the basic concepts underlying the good practice recommendations would apply to a continuous variable, the details of the accuracy assessment methodology (cf. Riemann, Wilson, Lister, & Parks, 2010) and area estimation would likely be considerably different from the methods presented herein.

4. Analysis

The analysis protocol specifies the measures to be used to express accuracy and class area as well as the procedures to estimate the selected measures from the sample data. In the context of studies of land change, there are two key objectives of the analysis: 1) accuracy assessment of the change classification, and 2) estimation of area of change. The confusion or error matrix (hereafter noted as the error matrix) plays a central role in meeting both the accuracy assessment and area estimation objectives (Foody, 2013; Stehman, 2013).

4.1. The error matrix

The error matrix is a simple cross-tabulation of the class labels allocated by the classification of the remotely sensed data against the reference data for the sample sites. The error matrix organizes the acquired sample data in a way that summarizes key results and aids the quantification of accuracy and area. The main diagonal of the error matrix highlights correct classifications while the off-diagonal elements show omission and commission errors. The cell entries and marginal values of the error matrix are fundamental to both accuracy assessment and area estimation. Table 4 illustrates a four-class example error matrix of the type often used in studies of land change.

The rows of the error matrix represent the labels shown in a map derived from the classification of the remote sensing data and the columns represent the labels depicted in the reference data. This layout is not a universal requirement and some may wish to reverse the contents of the rows and columns. In the matrix, p_{ij} represents the proportion of area for the population that has map class i and reference class j , where “population” is defined as the full region of interest, and p_{ij} is therefore the value that would result if a census of the population was obtained (i.e., complete coverage reference classification).

Accuracy parameters derived from a population error matrix of q classes include overall accuracy

$$O = \sum_{j=1}^q p_{jj} \quad (1)$$

user's accuracy of class i (the proportion of the area mapped as class i that has reference class i)

$$U_i = p_{ii}/p_{i\cdot} \quad (2)$$

or its complementary measure, commission error of class i , $1 - p_{ii}/p_{i\cdot}$, and producer's accuracy of class j (the proportion of the area of reference class j that is mapped as class j),

$$P_j = p_{jj}/p_{\cdot j} \quad (3)$$

or its complementary measure, omission error of class j , $1 - p_{jj}/p_{\cdot j}$. A variety of other measures of accuracy has been used in remote sensing (Liu, Frazier, & Kumar, 2007). A commonly used measure is the kappa coefficient of agreement (Congalton & Green, 2009). The problems associated with kappa include but are not limited to: 1) the correction for hypothetical chance agreement produces a measure that is not descriptive of the accuracy a user of the map would encounter (kappa would underestimate the probability that a randomly selected pixel is correctly

classified); 2) the correction for chance agreement used in the common formulation of kappa is based on an assumption of random chance that is not reasonable because it uses the map marginal proportions of area in the definition of chance agreement and these proportions are clearly not simply random; and 3) kappa is highly correlated with overall accuracy so reporting kappa is redundant with overall accuracy.” (Foody, 1992; Liu et al., 2007; Pontius & Millones, 2011; Stehman, 1997). Consistent with the recommendation in Strahler et al. (2006) the use of kappa is strongly discouraged as, despite its widespread use, it actually does not serve a useful role in accuracy assessment or area estimation.

4.2. General principles of estimation for good practice

The analysis protocol is designed to achieve the objectives of estimating accuracy and area from the sample data. Analysis thus requires statistical inference as the underlying scientific support for generalizing from the sample data to the population parameters and for quantifying uncertainty of the sample-based estimators. We recommend design-based inference (Särndal, Swensson, & Wretman, 1992) as the framework within which estimation is conducted. A fundamental tenet of design-based inference is that the specific estimators for accuracy, area, and the variances of these estimators depend on the sampling design implemented; different estimators are appropriate for different sampling designs. Therefore, it is essential that only unbiased or consistent estimators should be used. In practical terms, this means that only formulas for estimating parameters and variances that account for the inclusion probabilities associated with the sampling design implemented should be used. All recommended good practice estimators meet this condition, but the versions of the estimators presented are usually forms where the individual inclusion probabilities do not appear explicitly.

4.3. Estimating accuracy

The cell entries of the population error matrix and the parameters derived from it must be estimated from a sample. Suppose the sample-based estimator of p_{ij} is denoted as $\hat{p}_{ij}$. Once $\hat{p}_{ij}$ is available for each element of the error matrix, parameters can be estimated by substituting $\hat{p}_{ij}$ for p_{ij} in the formulas for the parameters. Accordingly, the error matrix should be reported in terms of these estimated area proportions, $\hat{p}_{ij}$, and not in terms of sample counts, n_{ij} . The specific formula for estimating p_{ij} depends on the sampling design used. For equal probability sampling designs (e.g., simple random and systematic sampling) and for stratified random sampling in which the strata correspond to the map classes,

$$\hat{p}_{ij} = W_i \frac{n_{ij}}{n_i} \quad (4)$$

where W_i is the proportion of area mapped as class i . For simple random and systematic sampling, Eq. (4) is a poststratified estimator of p_{ij} (Card, 1982) and for these sampling designs the poststratified estimator is recommended because it will have better precision than the estimators commonly used (cf. Stehman & Foody, 2009). Substituting $\hat{p}_{ij}$ of Eq. (4) into Eqs. (1)–(3) yields estimators of overall, user's, and producer's accuracies. These formulas are simpler special cases of a more general estimation approach described in Strahler et al. (2006, Eq. (3.1)).

The sampling variability associated with the accuracy estimates should be quantified by reporting standard errors. The variance estimators are provided below, and taking the square root of the estimated variance results in the standard error of the estimator. For overall accuracy, the estimated variance is

$$\hat{V}(\hat{O}) = \sum_{i=1}^q W_i^2 \hat{U}_i (1 - \hat{U}_i) / (n_i - 1). \quad (5)$$

For user's accuracy of map class i , the estimated variance is

$$\hat{V}(\hat{U}_i) = \hat{U}_i(1 - \hat{U}_i)/(n_i - 1). \quad (6)$$

For producer's accuracy of reference class $j = k$, the estimated variance is

$$\hat{V}(\hat{P}_j) = \frac{1}{\hat{N}_j^2} \left[\frac{N_j^2(1 - \hat{P}_j)^2 \hat{U}_j(1 - \hat{U}_j)}{n_j - 1} + \hat{P}_j^2 \sum_{i \neq j} N_i^2 \frac{n_{ij}}{n_i} \left(1 - \frac{n_{ij}}{n_i}\right) / (n_i - 1) \right] \quad (7)$$

where $\hat{N}_j = \sum_{i=1}^q \frac{N_i}{n_i} n_{ij}$ is the estimated marginal total number of pixels of reference class j , N_j is the marginal total of map class j and n_j is the total number of sample units in map class j . These are the usual variance estimators applied to the stratified sampling, and the estimators would be viewed as poststratified variance estimators for simple random and systematic sampling. For systematic sampling, the variance estimators are approximations that usually result in overestimation of variance. These variance estimators are also based on assumptions that the assessment unit for the response design is a pixel and each pixel has a hard classification for the map and a hard classification for the reference data. The variance estimators would not apply to a polygon assessment unit or to a mixed pixel situation.

4.4. Estimating area

The error matrix also provides the basis for estimating the area of classes such as those representing change and no-change. The population error matrix (Table 4) provides two different approaches for estimating the proportion of area. Suppose we are interested in estimating the proportion of area of class k . The row and column totals are the sums of the p_{ij} values in the respective rows and columns. Thus, the row total $p_{k\cdot}$ represents the proportion of area mapped as class k (e.g., if k is a change class such as forest loss then $p_{k\cdot}$ is the proportion of area mapped as forest loss) and the column total $p_{\cdot k}$ represents the proportion of area of class k as determined from the reference classification (e.g., $p_{\cdot k}$ would be the proportion of area of forest loss as determined from the reference classification).

The two area proportion parameters for class k (i.e., $p_{k\cdot}$ and $p_{\cdot k}$) are unlikely to have the same value, so a decision arises as to which parameter should be the focus. Once a change map is complete, $p_{k\cdot}$ is known, but because the reference classification is available only for a sample, $p_{\cdot k}$ must be estimated from the sample. Consequently, the need to estimate $p_{\cdot k}$ introduces uncertainty in the form of sampling variability, whereas $p_{k\cdot}$ is not subject to sampling variability (Stehman, 2005). The map-based parameter $p_{k\cdot}$ is known with certainty but likely biased because of classification error. Conversely, $p_{\cdot k}$ is determined from the reference classification. Therefore, $p_{\cdot k}$ should have smaller bias than $p_{k\cdot}$ (i.e., the bias attributable to reference data error is smaller than the bias attributable to map classification error). The "good practice" guidelines are founded on the premise that the reference classification is superior in quality relative to the map classification and that the sampling design implemented yields estimates with small standard errors. Consequently, we recommend that area estimation should be based on $p_{\cdot k}$, the proportion of area derived from the reference classification.

A variety of estimators has been proposed for estimating $p_{\cdot k}$ from the error matrix. For any sampling design and response design leading to an estimated error matrix with p_{ij} in terms of proportion of area, a direct estimator of the proportion of area of class k is

$$\hat{p}_{\cdot k} = \sum_{i=1}^q \hat{p}_{ik}. \quad (8)$$

This estimator is simply the sum of the estimated area proportions of class k as determined from the reference classification (i.e., the sum of column k of the estimated error matrix). If the sampling design is simple

random, systematic, or stratified random with the map classes defined as the strata, Eq. (8) would be computed using $\hat{p}_{ij} = W_i \frac{n_{ij}}{n_i}$ leading to the often used special case estimator

$$\hat{p}_{\cdot k} = \sum_{i=1}^q W_i \frac{n_{ik}}{n_i}. \quad (9)$$

This estimator is a poststratified estimator for simple random and systematic sampling, and it is the direct stratified estimator of $p_{\cdot k}$ for stratified random sampling when the map classes are the strata. For these sampling designs, the stratified estimator (Eq. (9)) generally has better precision than a variety of alternative estimators of area (Stehman, 2013) and consequently the stratified estimator is recommended.

For the stratified estimator of proportion of area (Eq. (9)), the standard error is estimated by

$$S(\hat{p}_{\cdot k}) = \sqrt{\sum_i W_i^2 \frac{n_{ik}}{n_i} \left(1 - \frac{n_{ik}}{n_i}\right)} = \sqrt{\sum_i W_i \frac{\hat{p}_{ik} - \hat{p}_{\cdot k}^2}{n_i - 1}} \quad (10)$$

where n_{ik} is the sample count at cell (i, k) in the error matrix, W_i is the area proportion of map class i , $\hat{p}_{ik} = W_i \frac{n_{ik}}{n_i}$ and the summation is over the q classes. For systematic sampling, Eq. (10) is an approximation that is typically an overestimate for the actual standard error of systematic sampling. The estimated area of class k is $\hat{A}_k = A \times \hat{p}_{\cdot k}$, where A is the total map area. The standard error of the estimated area is given by

$$S(\hat{A}_k) = A \times S(\hat{p}_{\cdot k}). \quad (11)$$

An approximate 95% confidence interval is obtained as $\hat{A}_k \pm 1.96 \times S(\hat{A}_k)$.

5. Example of good practices: estimating area and assessing accuracy of forest change

The following hypothetical example illustrates the workflow of assessing accuracy of a forest change map and estimating area. Consider a change map for 2000 to 2010 consisting of two change classes and two stable classes: deforestation, forest gain, stable forest and stable non-forest. The map was produced by supervised classification of data from Landsat ETM+ with the objective of estimating the gross rates of forest loss and gain. The first step in the assessment was to visually inspect the change map and identify obvious errors by comparing the classified results to the Landsat data of 2000 and 2010. Misclassified regions were relabeled before proceeding to the rigorous evaluation of the map. After obvious errors were removed, the areas of the map classes were 200,000 Landsat pixels (18,000 ha) of deforestation, 150,000 pixels (13,500 ha) of forest gain, 3,200,000 pixels (288,000 ha) of stable forest, and 6,450,000 pixels (580,500 ha) of stable non-forest. The two change classes thus occupy 3.5% of the total map area. The accuracy assessment was designed for the objectives of estimating overall and class-specific accuracies, areas of the individual classes (as determined by the reference classification), and confidence intervals for each accuracy and area parameter. The spatial assessment unit in this example is a Landsat pixel (30 m $\times$ 30 m).

5.1. Sampling design

A stratified random sampling design with the four map classes as strata adheres to the recommended practices outlined in Section 2 and satisfies the accuracy assessment and area estimation objectives. In the next two subsections, we present sample size and sample allocation planning calculations for the stratified design. Sample size planning is an inexact science because it is dependent on accuracy and area information that must be speculative prior to conducting the actual accuracy

assessment. Nevertheless, these planning calculations can provide informative insight into the choices of sample size and sample allocation to strata.

5.1.1. Determining the sample size

For simple random sampling and targeting overall accuracy as the estimation objective, Cochran (1977, Eq. (4.2)) suggests using a sample size of

$$n = \frac{z^2 O(1-O)}{d^2} \quad (12)$$

where O is the overall accuracy expressed as a proportion, z is a percentile from the standard normal distribution ($z = 1.96$ for a 95% confidence interval, $z = 1.645$ for a 90% confidence interval), and d is the desired half-width of the confidence interval of O . Eq. (12) provides a starting point for assessing sample size for the limited scope of estimating overall accuracy.

For stratified random sampling, Cochran (1977, Eq. (5.25)) provides the following sample size formula (the cost of sampling each stratum is assumed the same):

$$n = \frac{(\sum W_i S_i)^2}{[s(\hat{O})]^2 + (1/N) \sum W_i S_i^2} \approx \left(\frac{\sum W_i S_i}{s(\hat{O})} \right)^2 \quad (13)$$

where N = number of units in the ROI, $s(\hat{O})$ is the standard error of the estimated overall accuracy that we would like to achieve, W_i is the mapped proportion of area of class i , and S_i is the standard deviation of stratum i , $S_i = \sqrt{U_i(1-U_i)}$ (Cochran, 1977, Eq. (5.55)). Because N is typically large (e.g., over 10 million pixels in this example), the second term in the denominator of Eq. (13) can be ignored. We specify a target standard error for overall accuracy of 0.01. Suppose from past experience with similar change mapping efforts we know that errors of commission are relatively common for the change classes while the stable classes are more accurate (e.g., Olofsson et al., 2010, 2011). Consequently, we conjecture that user's accuracies of the two change classes will be 0.70 for deforestation and 0.60 for forest gain, and user's accuracies of the stable classes will be 0.90 for stable forest and 0.95 for stable non-forest. The resulting sample size from Eq. (13) is $n = 641$. These sample size calculations should be repeated for a variety of choices of $s(\hat{O})$ and U_i before reaching a final decision.

5.1.2. Determine sample allocation to strata

Once the overall sample size is chosen, we determine the allocation of the sample to strata. It is important that the sample size allocation results in precise estimates of accuracy and area. Stehman (2012) identifies four different approaches to sample allocation: proportional, equal, optimal and power allocation. In proportional allocation, the sample size per map class is proportional to the relative area of the map class. In this example, and which is usually the case when mapping land change, the mapped areas of change are small relative to other classes so proportional allocation will lead to small sample sizes in the rare classes (unless n is very large) and imprecise estimates of user's accuracy for these rare classes. Allocating an equal sample size to all strata targets estimation of user's accuracy of each map class but equal allocation is not optimized for estimating area and overall accuracy. Neyman optimal allocation (Cochran, 1977) can be used to minimize the variance of the estimator of overall accuracy or the estimator of area, but optimal allocation becomes difficult to implement when multiple estimation objectives are of interest as will be the case when estimating accuracy and area of several land-cover classes or land-cover change types.

We suggest the following simplified approach to sample size allocation. Allocate a sample size of 50–100 for each change strata using the variance estimator for user's accuracy (Eq. (6)) to decide the sample

size needed to achieve certain standard errors for the assumed estimated user's accuracy for that class. A small overall sample size might allow for only 50 sample units per rare class stratum. Suppose that $n-r$ sample units remain after a sample size of r units has been allocated to the rare class strata. The sample size of $n-r$ is then allocated proportionally to the area of each remaining stratum. The anticipated estimated variances can then be computed (based on the sample size allocation) for user's and overall accuracy and area using Eqs. (5), (6) and (10). The sample size allocation process can be iterated until an allocation is found that yields satisfactory anticipated standard errors for the key accuracy and area estimates. The effect of the choice of sample allocation will be observed in the standard errors of the estimates, however, a poor allocation of sample size to strata will not result in biased estimators.

In this example, we know the mapped areas of the four map classes (W_i), we have conjectured values of user's accuracies and standard errors of the strata, and we have estimated a total sample size of 641 (Table 5). The resulting sample sizes for proportional and equal allocation are shown in Table 5. As described above, neither of these is optimal and we want to find a compromise between the two. We start by allocating 100 sample units each to the change classes and then allocate the remainder of the sample size proportionally to the stable classes. This gives the allocation in column "Alloc1". Since the recommendation is to allocate between 50 and 100 sample units in the change strata, we introduce two additional allocations with 75 and 50 sample units in the change strata, respectively ("Alloc2" and "Alloc3"). To determine which of these allocations to use, we need to examine the standard errors of the estimated user's accuracy, estimated overall accuracy, and estimated areas using Eqs. (5), (6) and (10).

It is necessary to speculate the outcome of the accuracy assessment to compute the anticipated standard errors for each sample allocation considered. The hypothesized error matrix in Table 6 reflects the anticipated outcome that the change classes will be rare and have lower class-specific accuracies than the two stable classes. The population error matrix was also constructed to yield the hypothesized accuracies input into the sample size planning calculations of the previous section. When creating the hypothesized error matrix used for sample size and sample allocation planning, we should draw upon any past experience for insight into the accuracy of the map to be produced.

Table 7 shows the standard errors of the user's and overall accuracies and estimated areas of both deforestation and stable forest for each of the five sample allocations in Table 5 and the hypothetical population error matrix of Table 6. No single allocation is best for all estimation objectives, so a choice among competing objectives is necessary. The emphasis on prioritizing objectives during the planning stage (Section 2) becomes particularly relevant to the decision of sample allocation because different allocations favor different estimation objectives. For example, equal allocation gives the smallest standard error of the user's accuracy of deforestation but a high standard error of the estimated area of deforestation. Proportional allocation will result in smaller standard errors of overall accuracy and area of stable forest but the standard error for estimated user's accuracy of deforestation is two to four times larger than the corresponding standard errors for other sample allocations. In this case, "Alloc1–3" provide allocations that generate relatively small standard errors for the different estimates. We will choose "Alloc2" with 75 sample units in the two change classes.

Table 5

Information needed to decide allocation of sample size to strata. The information includes the mapped area proportions (W_i), conjectured values of user's accuracies (U_i) and standard deviations (S_i) of the strata. Columns 5–9 contain five different allocations.

Strata (i)	W_i	U_i	S_i	Equal	Alloc1	Alloc2	Alloc3	Prop
1 Deforestation	0.020	0.700	0.458	160	100	75	50	13
2 Forest gain	0.015	0.600	0.490	160	100	75	50	10
3 Stable forest	0.320	0.900	0.300	160	149	165	182	205
4 Stable non-forest	0.645	0.950	0.218	160	292	325	358	413

Table 6

Hypothetical population error matrix expressed in terms of proportion of area (see Section 4) used for sample size and sample allocation planning calculations.

		Reference				
		Deforestation	Forest gain	Stable forest	Stable non-forest	Total (W_i)
Map	Deforestation	0.014	0	0.003	0.003	0.020
	Forest gain	0	0.009	0.003	0.003	0.015
	Stable forest	0.002	0	0.288	0.030	0.320
	Stable non-forest	0.004	0.002	0.025	0.614	0.645
	Total	0.020	0.011	0.319	0.650	1

5.2. Estimating accuracy, area and confidence intervals

To create the reference classification for labeling each sample unit, a combination of Landsat data from the USGS open archive together with GoogleEarth™ provides a source of cost free reference data. Our hypothetical map was produced using Landsat, and the good practice recommendations stipulate that if using the same data for creation of both the map and reference classifications, the process of creating the latter should be of higher quality than the map-making process. The process of labeling the sample units thus has to be more accurate than supervised classification. A manual inspection by three analysts of each of the sample units using a set of Landsat images together with GoogleEarth™ imagery acquired around the same time as the images used to make the map is assumed to be a more accurate process than supervised classification. The error matrix resulting from this response design and sample is presented in terms of the sample counts displayed in Table 8, and the computations for the accuracy and area estimates are detailed in the following two subsections.

5.2.1. Estimating accuracy

Because the sampling design is stratified random using the map classes as strata, the cell entries of the error matrix are estimated using Eq. (4).

We can now estimate user's accuracy $\hat{U}_i = \frac{\hat{p}_{ii}}{\hat{p}_{i\cdot}}$; producer's accuracy $\hat{P}_j = \frac{\hat{p}_{jj}}{\hat{p}_{\cdot j}}$; and overall accuracy $\hat{O} = \sum_{j=1}^q \hat{p}_{jj}$ using the estimated area proportions. Variances for these accuracy measures are estimated using Eqs. (5)–(7). 95% confidence intervals are estimated as $\pm 1.96 \sqrt{\hat{V}(\hat{U}_i)}$ (replace $\hat{U}_i$ with $\hat{P}_j$ and $\hat{O}$ for the producer's and overall accuracies). In this case, the estimated user's accuracy ($\pm 95\%$ confidence interval) is 0.88 ± 0.07 for deforestation, 0.73 ± 0.10 for forest gain, 0.93 ± 0.04 for stable forest, and 0.96 ± 0.02 for stable non-forest. The estimated producer's accuracy is 0.75 ± 0.21 for deforestation, 0.85 ± 0.23 for forest gain, 0.93 ± 0.03 for stable forest, and 0.96 ± 0.01 for stable non-forest. The estimated overall accuracy is 0.95 ± 0.02 .

5.2.2. Estimating area and uncertainty

The next step is to use the estimated area proportions in Table 9 to estimate the area of each class. The row totals of the error matrix in

Table 7

Standard errors of selected accuracy and area estimates for different sample size allocations to strata (Table 5) and the hypothetical population error matrix (Table 6). Standard errors are shown for estimated overall accuracy, estimated user's accuracy for the rare class deforestation ($i = 1$) and the common class stable forest ($i = 3$), and estimated area (in units of hectares) of deforestation and area of stable forest.

Allocation	$s(\hat{O})$	$s(\hat{U}_1)$	$s(\hat{U}_3)$	$s(\hat{A}_1)$	$s(\hat{A}_3)$
Equal	0.013	0.036	0.024	4035	11,306
Alloc1	0.011	0.046	0.025	3307	9744
Alloc2	0.011	0.053	0.023	3138	9270
Alloc3	0.010	0.065	0.022	3125	8860
Proportional	0.010	0.132	0.021	3600	8614

Table 9 give the mapped area proportions (which are also given by W_i) while the column totals give the estimated area proportions according to the reference data. Multiplying the latter by the total map area gives the stratified area estimate of each class according to the reference data. For example, the estimated area of deforestation according to the reference data is $\hat{A}_1 = \hat{p}_{\cdot 1} \times A_{tot} = 0.024 \times 10,000,000 \text{ pixels} = 235,086 \text{ pixels} = 21,158 \text{ ha}$. The mapped area of deforestation ($A_{m,1}$) of 200,000 pixels was thus underestimated by 35,086 pixels or 3158 ha.

The second step is to estimate a confidence interval for the area of each class. From Eq. (10), $S(\hat{p}_{\cdot 1}) = 0.0035$ and the standard error for the estimated area of forest loss is $S(\hat{A}_1) = S(\hat{p}_{\cdot 1}) \times A_{tot} = 0.0035 \times 10,000,000 = 34,097 \text{ pixels}$. The margin of error of the confidence interval is $1.96 \times 34,097 = 68,418 \text{ pixels} = 6158 \text{ ha}$. We have thus estimated the area of deforestation with a 95% confidence interval: $21,158 \pm 6158 \text{ ha}$. The area estimate with a 95% confidence interval of the forest gain class is $11,686 \pm 3756 \text{ ha}$; stable forest is $285,770 \pm 15,510 \text{ ha}$ and stable non-forest $581,386 \pm 16,282 \text{ ha}$.

This example has illustrated the workflow of assessing accuracy, and estimating area and confidence intervals of area of the classes of a change map. While this is fairly straightforward once the error matrix has been constructed, the example highlights the need to consider different objectives when designing the sample.

A tool for estimating unbiased accuracy measures and areas with 95% confidence intervals can be downloaded from www.people.bu.edu/olofsson/ (click 'Research' > 'Accuracy/Uncertainty'). The tool is implemented in Matlab™.

6. Summary

Conducting an accuracy assessment of a land change map serves multiple purposes. In addition to the obvious purpose of quantifying the accuracy of the map, the reference sample serves as the basis of estimates of area of each class where area is defined by the reference classification. The accuracy assessment sample data also contribute to estimates of uncertainty of the area estimates. Without an accuracy assessment, there is no way to communicate map quality in a quantitative and meaningful fashion. We acknowledge that there is no singular "best" approach and the recommendations provided do not preclude the existence of other acceptable practices. However, by following the "good practice" recommendations presented by this paper, scientific credibility of the accuracy and area estimates is ensured. The "good practice" recommendations are summarized as follows, organized by the three major components of the accuracy assessment methodology, the sampling design, response design, and analysis.

6.1. General

- Visually inspect the map and correct obvious errors before conducting the accuracy assessment.
- Accuracy and area estimates will be determined from a classification (i.e., the reference classification) that is of higher quality than the land change map being evaluated.
- A sampling approach is needed because the cost of obtaining the reference classification for the entire region of interest will be prohibitive.

Table 8Description of sample data as an error matrix of sample counts, n_{ij} (see Table 9 for recommended estimated error matrix used to report accuracy results).

		Reference					
		Deforestation	Forest gain	Stable forest	Stable non-forest	Total	$A_{m,i}$ [pixels]
Map	Deforestation	66	0	5	4	75	200,000
	Forest gain	0	55	8	12	75	150,000
	Stable forest	1	0	153	11	165	3,200,000
	Stable non-forest	2	1	9	313	325	6,450,000
	Total	69	56	175	340	640	10,000,000

- The sample used for accuracy assessment and area estimation is separate from (independent of) the data used to train or develop the classification.

6.2. Sampling design

- Implement a probability sampling design to provide a rigorous foundation via design-based sampling inference.
- Document and quantify any deviations from the probability sampling protocol.
- Choose a sampling design on the basis of specified accuracy objectives and prioritized desirable design criteria.
- Sampling design guidelines.
 - Stratify by map class to reduce standard errors of class-specific accuracy estimates.
 - If resources are adequate, stratify by subregions to reduce standard errors of subregion-specific estimates.
 - Use cluster sampling if it provides a substantial cost savings or if the objectives require a cluster unit for the assessment.
 - Both simple random and systemic selection protocols are acceptable options.
- The recommended allocation of sample size to strata (assuming the map classes are the strata) is to increase the sample size for rare change classes to achieve an acceptable standard error for estimated user's accuracies and to allocate the remaining sample size roughly proportional to the area occupied by the common classes.
- Use sample size and optimal allocation planning calculations as a guide to decisions on total sample size and sample allocation.
- Evaluate the potential outcome of sample size and sample allocation decisions on the standard errors of accuracy and area estimates for hypothetical error matrices based on the anticipated accuracy of the map.
- Stratified random sampling using the map classification to define strata is a simple, but generally applicable design that will typically satisfy most accuracy and area estimation objectives and desirable design criteria.

6.3. Response design

- Reference data should be of higher quality than the data used for creating the map, or if using the same source, the process of creating the reference classification should be more accurate than the process of creating the map.

- High overhead cost may eliminate field visits as a source of reference data.
- The reference data should provide sufficient temporal representation consistent with the change period of the map.
- Data from the Landsat open archive in combination with high spatial resolution imagery provide a low-cost and often useful source of reference data (national photograph archives, satellite photo archives (e.g., Kompsat), and the collections available through Google Earth™ are possible high resolution imagery sources).
- Specify protocols for accounting for uncertainty in assigning the reference classifications.
- Assign each sample unit a primary and secondary label (secondary not required if there is highly confidence in the primary label).
- Include an interpreter specified confidence for each reference label (e.g., high, medium, or low confidence).
- Implement protocols to ensure consistency among individual interpreters or teams of interpreters.
- Specify a protocol for defining agreement between the map and reference classifications that will lead to an error matrix expressed in terms of proportion of area.

6.4. Analysis

- Report the error matrix in terms of estimated area proportions.
- Report the area (or proportion of area) of each class as determined from the map.
- Report user's accuracy (or commission error), producer's accuracy (or omission error), and overall accuracy (Eqs. (1)–(3)).
- Avoid use of the kappa coefficient of agreement for reporting accuracy of land change maps.
- Estimate the area of each class according to the classification determined from the reference data.
- Use estimators of accuracy and area that are unbiased or consistent.
- For simple random, systematic, and stratified random sampling when the map classes are defined as strata, use stratified estimators of accuracy (Eqs. (5)–(7)) and a stratified estimator of area (Eq. (9)).
- Quantify sampling variability of the accuracy and area estimates by reporting standard errors or confidence intervals.
- Use design-based inference to define estimator properties and to quantify uncertainty.
- Assess the impact of reference data uncertainty on the accuracy and area estimates.

Table 9

The error matrix in Table 8 populated by estimated proportions of area.

		Reference					
		Deforestation	Forest gain	Stable forest	Stable non-forest	Total (W_i)	$A_{m,i}$ [pixels]
Map	Deforestation	0.0176	0	0.0013	0.0011	0.020	200,000
	Forest gain	0	0.0110	0.0016	0.0024	0.015	150,000
	Stable forest	0.0019	0	0.2967	0.0213	0.320	3,200,000
	Stable non-forest	0.0040	0.0020	0.0179	0.6212	0.645	6,450,000
	Total	0.0235	0.0130	0.3175	0.6460	1	10,000,000

The recommendations provided are intended to serve as guidelines for choosing from among options of sampling design, response design, and analysis that will yield rigorous and defensible accuracy and area estimates. But good practice is not static. As improvements in technology become available and new methods are developed, good practice recommendations will evolve over time. Also, as practical experience accumulates with using new technology and methodologies, good practice recommendations will be further amended to provide even more efficient yet still rigorous methods to estimate accuracy and area of land change.

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